Boundary Compensation Origin V: Simplicial Z_2 Shadow-Gauge Ensembles Spectral Wilson Weights, Structural Inhomogeneous-Bond Couplings, and Finite-Graph Confinement Diagnostics

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Abstract

This review manuscript, updated after the statistical-bridge review, develops the next structural layer of the BC-Origin programme after the construction of non-factorizable edge holonomy in BC-Origin III and lifted phase-flow shadow kinematics in BC-Origin IV. The central question is whether the holonomy data of multi-shadow geometry can generate gauge-type cycle weights without inserting a Wilson action by hand.

Let $\mathcal{K} = (V, E, F_2)$ be a finite hidden-index two-dimensional simplicial complex, let $K_{ij} = K(n_i, n_j) \geq 0$ be a structural kernel on edges, and let $\varepsilon_{ij} \in \{-1, +1\}$ be a non-factorizable Z_2 edge holonomy. We define the off-diagonal holonomy-weighted structural adjacency operator

$$(A_\varepsilon)_{ii} = 0, \qquad (A_\varepsilon)_{ij} = \varepsilon_{ij} K_{ij}.$$

The first main result is the closed-walk trace identity

$$\mathrm{Tr}(A_\varepsilon^m) = \sum_{C:\ |C|=m} \Omega_C K_C,$$

where the sum runs over closed walks of length m, Ω_C is the product of edge holonomies along the walk, and K_C is the product of structural kernels along the same walk. In particular, for triangular cycles,

$$\mathrm{Tr}(A_\varepsilon^3) = 6 \sum_{\triangle} \Omega_\triangle K_{ij} K_{jk} K_{ki}.$$

Thus Wilson-type Z_2 cycle products are not externally imposed; they occur as gauge-invariant spectral trace terms of the multi-shadow holonomy operator.

The second main step is a statistical bridge. Version v0.1.3 implements the full statistical-bridge patch requested in review: the trace is not identified with an action. Instead, a finite simplicial gauge-weight ensemble is defined by

$$Z_3(\gamma) = \sum_\varepsilon \exp\left(\gamma_3 \,\mathrm{Tr}(A_\varepsilon^3)\right),$$

or by multi-trace extensions. This converts spectral cycle observables into Wilson-like statistical weights while keeping the structural dependence in the kernel products K_C. The resulting ensemble is finite-dimensional, simplicial-complex-based, and toy-model-level. It is not claimed to be physical lattice quantum field theory, QCD, a continuum gauge theory, or an empirical confinement model. Its controlled contribution is narrower but nontrivial: it shows how Z_2 gauge-cycle weights and inhomogeneous structural couplings arise from BC-Origin spectral geometry.

Revision note for v0.1.3. This review manuscript implements the full reviewer statistical-bridge patch. The spectral trace is explicitly treated as a gauge-invariant observable / trace functional of one fixed holonomy configuration, not as an action by itself. The pure gauge-cycle derivation uses the off-diagonal holonomy adjacency operator A_ε, not the full shadow operator $D_\varepsilon = \Delta + \eta_D A_\varepsilon$, because the diagonal closure denominators in D_ε generate additional non-cycle contributions. The ensemble appears only after summing over edge-holonomy configurations:

$$Z_3(\gamma) = \sum_\varepsilon \exp\left(\gamma \operatorname{Tr}(A_\varepsilon^3)\right).$$

Using the triangle trace identity, this becomes a finite simplicial Z_2 shadow-gauge ensemble with structurally inhomogeneous Wilson-like couplings

$$\beta_\triangle = 6\gamma K_{ij} K_{jk} K_{ki}.$$

The revision also adds a strong-coupling/high-temperature form, simplicial two-cell terminology, deterministic inhomogeneous-bond terminology, finite-graph area-law diagnostics, and shadow-admissibility statistics. The non-claim boundary is preserved: no physical lattice QFT, QCD, continuum confinement theorem, or empirical gauge dynamics is asserted.

Contents

1 Introduction: from trace cycles to simplicial shadow-gauge ensembles

BC-Origin I introduced hidden oriented winding shadows and structural scale selection. BC-Origin II replaced free pairwise couplings by structural kernels $K(n_i, n_j)$. BC-Origin III introduced non-factorizable edge holonomy and genuine cycle frustration. BC-Origin IV introduced lifted phase flow and horizon events.

BC-Origin V asks a new question:

$$\text{edge holonomy} + \text{structural kernels} \quad \Longrightarrow \quad \text{spectral trace cycle weights?}$$

The answer is affirmative in a finite-dimensional and graph-theoretic sense. The closed spectral traces of the holonomy-weighted adjacency operator generate Wilson-cycle products automatically.

The key methodological distinction of the paper is

$$\mathrm{Tr}(A_\varepsilon^m) \neq \text{Wilson action by itself.}$$

Instead, $\mathrm{Tr}(A_\varepsilon^m)$ is a gauge-invariant trace functional that can be used to define a finite simplicial ensemble over holonomy configurations. This keeps the programme away from the category error of equating an observable of one fixed configuration with an action governing a statistical or path-integral sum over configurations.

Guiding chain. The controlled chain developed here is

$$\text{holonomy} \to \text{spectral trace cycles} \to \text{finite simplicial gauge-weight ensemble} \to \text{Wilson-loop diagnostics} \to \text{sha}$$

This is intentionally stronger than a sterile note about traces. It is a finite-dimensional spectral-gauge construction with an actual ensemble, Wilson-like cycle weights, loop observables and horizon-admissibility statistics. It is nevertheless narrower than a physical derivation of lattice quantum field theory.

2 Hidden-index simplicial complex and structural kernels

Let $\mathcal{K} = (V, E, F_2)$ be a finite two-dimensional simplicial complex whose vertices label hidden winding generators

$$h_i = (q_i, n_i), \qquad n_i \in \mathbb{Z} \setminus \{0\}.$$

A structural kernel is assigned to edges, and the declared triangular faces F_2 are the elementary two-cells used for Wilson-type factors:

$$K_{ij} = K(n_i, n_j), \qquad K_{ij} = K_{ji} \geq 0.$$

Typical examples include the Lorentzian index-space kernel

$$K_{ij} = \frac{\eta}{1 + (n_i - n_j)^2}$$

and exponential or finite-range kernels. The normalization η is global; it is not a pairwise fitted coupling.

Definition 1 (Hidden-index structural two-complex)**.** *A hidden-index structural two-complex is a tuple*

$$\mathcal{K} = (V, E, F_2, n, K),$$

where V is a finite vertex set, E is a finite unoriented edge set, F_2 is a declared set of filled triangular two-cells, $n : V \to \mathbb{Z} \setminus \{0\}$ assigns hidden winding labels, and $K : E \to \mathbb{R}_{\geq 0}$ assigns symmetric structural kernel values.

A triangle $\triangle = (ijk)$ contributes as an elementary Wilson-type face only when $\triangle \in F_2$. A graph-theoretic three-cycle not declared as an element of F_2 is merely a closed walk and is not automatically treated as a plaquette-like two-cell.

3 Z_2 edge holonomy and vertex gauge equivalence

Assign edge variables

$$\varepsilon_{ij} \in \{-1, +1\}, \qquad \varepsilon_{ij} = \varepsilon_{ji}.$$

A vertex gauge transformation is

$$\varepsilon_{ij} \mapsto g_i \varepsilon_{ij} g_j, \qquad g_i \in \{-1, +1\}.$$

For a closed cycle $C = (i_0 i_1 \cdots i_m = i_0)$, define

$$\Omega_C = \prod_{\ell=0}^{m-1} \varepsilon_{i_\ell i_{\ell+1}}.$$

Then Ω_C is invariant under vertex gauge transformations.

Proposition 1 (Cycle holonomy gauge invariance)**.** *For any closed cycle C, the cycle product Ω_C is unchanged by vertex gauge transformations $\varepsilon_{ij} \mapsto g_i \varepsilon_{ij} g_j$.*

Proof. Each vertex sign g_i appears exactly twice in the product around a closed cycle. Since $g_i^2 = 1$, all vertex factors cancel. $\square$

This recalls the BC-Origin III distinction:

$$\text{vertex-factorizable signs} \neq \text{non-factorizable edge holonomy}.$$

Only the latter can support genuine cycle-level frustration.

4 Holonomy-weighted structural adjacency operator

Define

$$(A_\varepsilon)_{ii} = 0,$$

$$(A_\varepsilon)_{ij} = \varepsilon_{ij} K_{ij} \quad \text{for } (ij) \in E,$$

and

$$(A_\varepsilon)_{ij} = 0 \quad \text{when } (ij) \notin E.$$

The full shadow operator may later be written as

$$D_\varepsilon(\lambda) = \Delta(\lambda) + \eta_D A_\varepsilon,$$

where $\Delta(\lambda) = \mathrm{diag}(d_i(\lambda))$ contains closure denominators. The pure gauge-cycle trace identities are cleanest for A_ε, because diagonal terms in D_ε produce additional non-cycle contributions.

Proposition 2 (Gauge covariance of A_ε). *Let $G_g = \mathrm{diag}(g_i)$ and let $\varepsilon^g_{ij} = g_i \varepsilon_{ij} g_j$. Then*

$$A_{\varepsilon^g} = G_g A_\varepsilon G_g.$$

Consequently, A_ε and A_{ε^g} have the same spectrum and the same trace powers.

Proof. The (i,j) entry of $G_g A_\varepsilon G_g$ is $g_i \varepsilon_{ij} K_{ij} g_j$, which equals $(A_{\varepsilon^g})_{ij}$. Similarity by $G_g = G_g^{-1}$ preserves the spectrum and trace powers. □

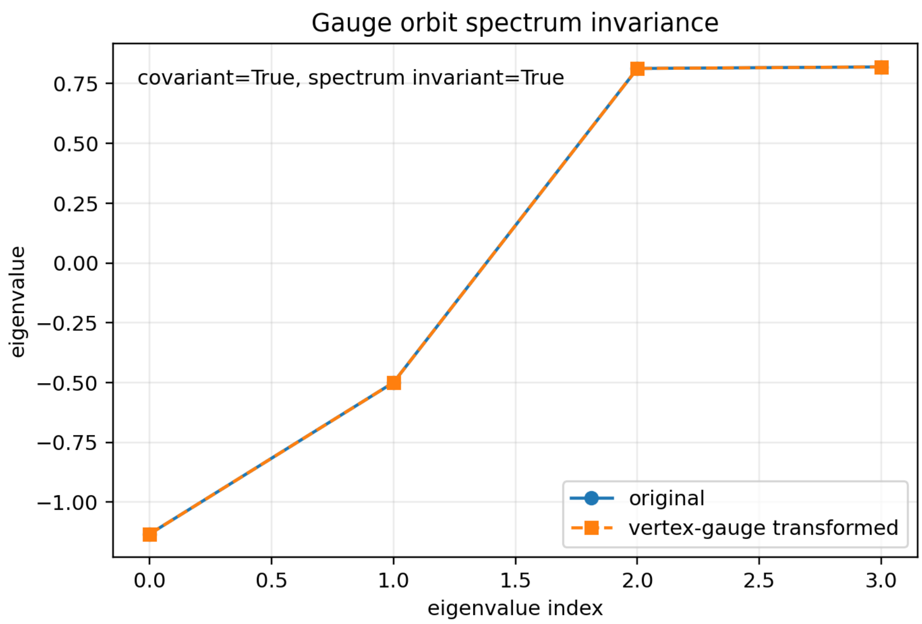

Figure 1: Gauge orbit spectrum invariance. A vertex sign redefinition changes edge signs but preserves the spectrum of the holonomy-weighted adjacency operator.

5 Main theorem: spectral traces generate closed holonomy walks

Theorem 1 (Closed-walk trace identity). *For the holonomy-weighted adjacency operator A_ε,*

$$\mathrm{Tr}(A_\varepsilon^m) = \sum_{i_0,\ldots,i_{m-1}} (A_\varepsilon)_{i_0 i_1} (A_\varepsilon)_{i_1 i_2} \cdots (A_\varepsilon)_{i_{m-1} i_0}.$$

Equivalently,

$$\mathrm{Tr}(A_\varepsilon^m) = \sum_{C:\ |C|=m} \Omega_C K_C,$$

where the sum runs over closed walks of length m,

$$\Omega_C = \prod_{(ij)\in C} \varepsilon_{ij},$$

and

$$K_C = \prod_{(ij)\in C} K_{ij}.$$

Thus every closed-walk term in the spectral trace carries a gauge-invariant holonomy factor.

Proof. Expand the diagonal entries of A_ε^m and sum over the starting vertex. Each monomial corresponds to a closed walk. Since each off-diagonal matrix entry factorizes as $\varepsilon_{ij} K_{ij}$, the monomial factorizes into the corresponding holonomy product Ω_C and kernel product K_C. □

This theorem is the first central result of BC-Origin V. It shows that Wilson-type cycle products are not decorative analogies; they are spectral trace coefficients of the BC-Origin holonomy-weighted structural adjacency operator.

6 Triangle Wilson-cycle corollary

For $m = 3$, only triangular closed walks contribute on a simple graph.

Corollary 1 (Triangle trace formula)**.** *For a simple graph,*

$$\mathrm{Tr}(A_\varepsilon^3) = 6 \sum_{i<j<k} \varepsilon_{ij}\varepsilon_{jk}\varepsilon_{ki} K_{ij} K_{jk} K_{ki}.$$

Equivalently,

$$\mathrm{Tr}(A_\varepsilon^3) = 6 \sum_{\triangle} \Omega_\triangle K_\triangle.$$

The factor 6 *counts the two orientations and three cyclic starting points of each triangle.*

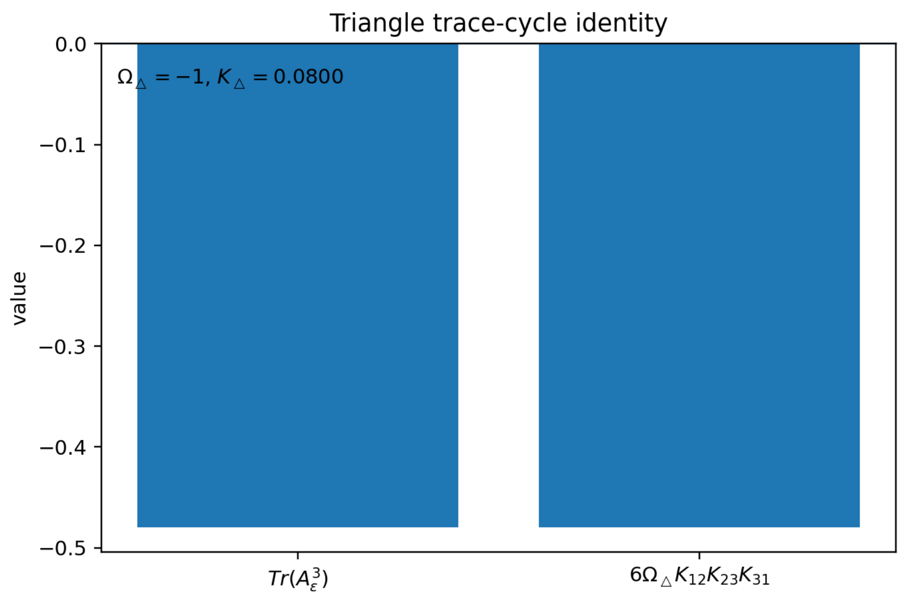

Figure 2: Triangle trace-cycle identity. The direct spectral trace $\mathrm{Tr}(A_\varepsilon^3)$ agrees with the Wilson-cycle expression $6\Omega_\triangle K_{12}K_{23}K_{31}$.

This is the minimal BC-Origin Wilson-cycle emergence statement.

7 The statistical bridge: from spectral trace observables to a finite Z_2 gauge-weight ensemble

BC-Origin III introduced non-factorizable edge holonomies

$$\varepsilon_{ij} \in \{-1, +1\}$$

on a finite hidden-index graph. Under a vertex gauge transformation

$$\varepsilon_{ij} \mapsto g_i \varepsilon_{ij} g_j, \qquad g_i \in \{-1, +1\},$$

cycle products

$$\Omega_C = \prod_{(ij)\in C} \varepsilon_{ij}$$

are gauge-invariant. BC-Origin V asks whether these gauge-invariant cycle products are merely imposed by hand, or whether they arise naturally from spectral invariants of the shadow operator.

To isolate the pure gauge-cycle part, define the off-diagonal holonomy-weighted structural adjacency operator

$$(A_\varepsilon)_{ii} = 0,$$

$$(A_\varepsilon)_{ij} = \varepsilon_{ij} K_{ij} \quad \text{for } (ij) \in E,$$

where $K_{ij} = K(n_i, n_j) \geq 0$ is the structural kernel inherited from the hidden winding-index geometry.

The full shadow operator used for admissibility diagnostics will later be written as

$$D_\varepsilon = \Delta + \eta_D A_\varepsilon,$$

where

$$\Delta = \mathrm{diag}(d_1, \ldots, d_N)$$

contains closure denominators. However, the clean Wilson-cycle trace identity belongs to A_ε, not to D_ε, because diagonal closure terms generate additional non-cycle contributions.

For the cubic trace,

$$\mathrm{Tr}(A_\varepsilon^3) = \sum_{i,j,k} (A_\varepsilon)_{ij} (A_\varepsilon)_{jk} (A_\varepsilon)_{ki}.$$

Substituting the structural edge form gives

$$(A_\varepsilon)_{ij} (A_\varepsilon)_{jk} (A_\varepsilon)_{ki} = (\varepsilon_{ij} K_{ij})(\varepsilon_{jk} K_{jk})(\varepsilon_{ki} K_{ki}).$$

Thus

$$(A_\varepsilon)_{ij} (A_\varepsilon)_{jk} (A_\varepsilon)_{ki} = \Omega_{ijk}\, K_{ij} K_{jk} K_{ki},$$

where

$$\Omega_{ijk} = \varepsilon_{ij}\varepsilon_{jk}\varepsilon_{ki}$$

is the Z_2 Wilson cycle around the triangle (ijk). Therefore, on a simple undirected triangular graph,

$$\mathrm{Tr}(A_\varepsilon^3) = 6 \sum_{\triangle} \Omega_\triangle K_\triangle,$$

with

$$K_\triangle = K_{ij} K_{jk} K_{ki}.$$

The factor 6 counts the two orientations and the three cyclic starting points of each triangle.

This establishes the first bridge:

$$\text{spectral trace cycle} \quad \Longrightarrow \quad Z_2 \text{ Wilson-cycle term with structural kernel weight.}$$

However, this trace is not yet an action. It is a gauge-invariant observable of one fixed holonomy configuration ε. A statistical or path-sum model appears only after summing over all edge-holonomy configurations.

We therefore define the finite graph partition function

$$Z_3(\gamma) = \sum_{\varepsilon \in \{-1,+1\}^E} \exp\left(\gamma\, \mathrm{Tr}(A_\varepsilon^3)\right).$$

Using the trace identity,

$$Z_3(\gamma) = \sum_{\varepsilon} \exp\left(6\gamma \sum_{\triangle} \Omega_\triangle K_\triangle\right).$$

Equivalently,

$$Z_3(\gamma) = \sum_{\varepsilon} \exp\left(\sum_{\triangle} \beta_\triangle \Omega_\triangle\right),$$

where the effective triangular Wilson weight is

$$\beta_\triangle = 6\gamma K_{ij} K_{jk} K_{ki}.$$

Thus the Wilson-like action is not inserted as a primitive postulate. It is generated by using the spectral trace functional of the holonomy-weighted structural adjacency operator as the energy functional of a finite Z_2 gauge-weight ensemble.

The global parameter γ is an ensemble-control parameter. It controls how strongly the statistical sum weights the cubic spectral trace of the holonomy adjacency operator. It is analogous to an inverse temperature or trace-weighting strength, but in BC-Origin it is not identified with physical temperature, physical time, or a fundamental gauge coupling.

The limiting readings are purely ensemble-theoretic. For $\gamma \to 0$, edge-holonomy configurations are nearly unweighted and the gauge background is maximally fluctuating. For large positive γ, configurations with larger structural trace contribution are exponentially favored, so the ensemble becomes more strongly organized by the declared face weights $K_\triangle$. The local deterministic inhomogeneous coupling structure remains determined by the structural kernel products $K_\triangle$.

Definition 2 (BC-Origin finite simplicial Z_2 shadow-gauge ensemble)**.** *Given a structural two-complex $\mathcal{K} = (V, E, F_2, n, K)$, define*

$$Z(\gamma) = \sum_{\varepsilon \in \{-1,+1\}^E} \exp\left(\sum_{\triangle \in F_2} \beta_\triangle \Omega_\triangle\right),$$

where

$$\beta_\triangle = 6\gamma K_\triangle.$$

This is the finite simplicial Z_2 shadow-gauge ensemble generated by the cubic trace of A_ε on the declared triangular two-cells.

This definition is intentionally finite-dimensional. It is not a continuum limit and not a physical path integral over spacetime gauge fields.

8 Strong-coupling / high-temperature form and Wilson-like local weights

For one triangle,

$$\exp(\beta_\triangle \Omega_\triangle) = \cosh(\beta_\triangle)\left[1 + \Omega_\triangle \tanh(\beta_\triangle)\right],$$

where

$$\beta_\triangle = 6\gamma K_{ij} K_{jk} K_{ki}.$$

Thus the effective Wilson-like triangle weight is structurally modulated:

$$u_\triangle = \tanh(6\gamma K_{ij} K_{jk} K_{ki}).$$

Unlike a uniform Wilson lattice action, BC-Origin produces inhomogeneous structural weights because K_{ij} depends on hidden winding-index geometry.

The parameter $u_\triangle$ is the natural finite-simplicial-complex strong-coupling/high-temperature expansion variable. Small $|u_\triangle|$ corresponds to a strongly fluctuating holonomy ensemble on the declared finite complex, while large $|u_\triangle|$ favors ordered triangle holonomies. In this manuscript such statements are finite-graph ensemble statements only; thermodynamic phase transitions require a graph-size limit and finite-size scaling, which are outside the claim set of v0.1.3.

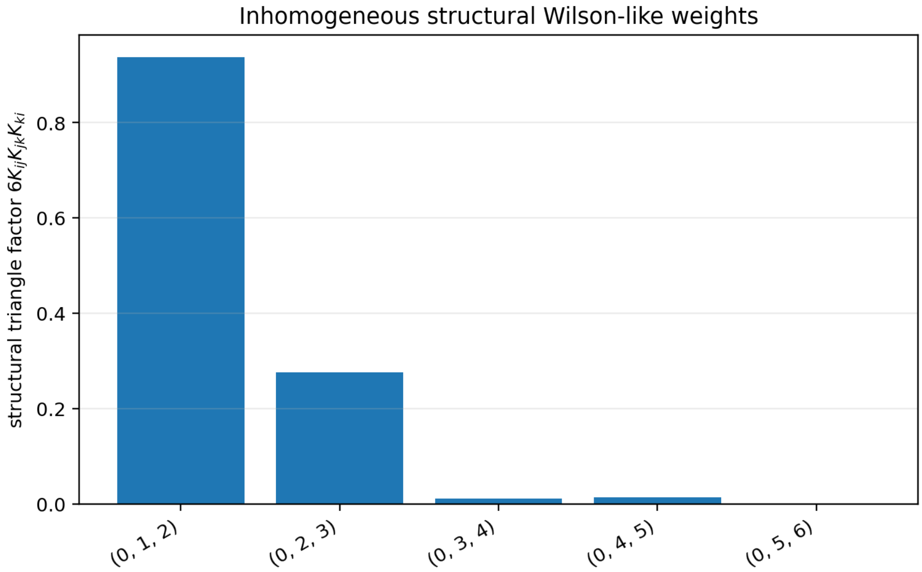

Figure 3: Structural inhomogeneity. Different triangles acquire different Wilson-like weights because the kernel products $K_{ij}K_{jk}K_{ki}$ depend on hidden winding-index geometry. No stochastic disorder average is used; these weights are deterministic once the hidden labels and kernel are fixed.

9 Finite simplicial interpretation: triangles as elementary 2-cells

The appearance of triangular cycles is not an error. It indicates that the natural finite structure is not necessarily a hypercubic lattice, but a graph or simplicial complex whose elementary closed 2-cells may be triangles.

If a 2-complex structure is declared, triangle holonomies can be interpreted as plaquette-like variables on triangular faces. The article does not claim a continuum limit, Regge calculus, dynamical triangulations, or quantum gravity. It only identifies the finite simplicial gauge-weight structure naturally generated by spectral trace cycles.

10 Deterministic inhomogeneous-bond structure

Because

$$\beta_\triangle = 6\gamma K_{ij}K_{jk}K_{ki},$$

different triangles generally have different weights. This produces a deterministic inhomogeneous-bond Z_2 gauge ensemble. No quenched disorder average is performed in BC-Origin V, and no probability measure over hidden labels is assumed. A stochastic disorder model would arise only after adding an explicit probability law on hidden-index data and averaging the appropriate thermodynamic quantities; that extension is outside the present manuscript.

11 Finite-graph crossover and confinement-like diagnostics

The ensemble supports observables analogous to standard lattice-gauge diagnostics, but only on the declared finite graph or finite simplicial complex. Define the average triangle holonomy

$$\langle \Omega_\triangle \rangle_\gamma = \frac{1}{|F_2|} \sum_{\triangle \in F_2} \langle \Omega_\triangle \rangle_\gamma,$$

and a finite-graph frustration energy

$$E_F(\gamma) = 1 - \langle \Omega_\triangle \rangle_\gamma.$$

The derivative or variance of this quantity under a sweep of γ may show a finite-size crossover or susceptibility peak. The paper does not call this a thermodynamic phase transition unless an appropriate sequence of graph refinements and scaling limit is supplied.

For closed cycles C one may compute Wilson-loop-like expectations $\langle \Omega_C \rangle_\gamma$ and compare $-\log |\langle \Omega_C \rangle_\gamma|$ with a declared enclosed triangle count. This is a finite-graph area-law diagnostic, not a proof of continuum confinement.

12 Coupling to shadow denominators

The observable shadow operator may be written

$$D_\varepsilon(\lambda) = \Delta(\lambda) + \eta_D A_\varepsilon,$$

where

$$\Delta(\lambda) = \mathrm{diag}(d_i(\lambda)).$$

The eigen-denominators $\lambda_a(D_\varepsilon)$ define local admissibility through

$$\lambda_{\min}(D_\varepsilon) > 0.$$

In the ensemble, one may define an admissibility fraction

$$P_{\mathrm{adm}}(\gamma) = \left\langle \mathbf{1}_{\lambda_{\min}(D_\varepsilon) > 0} \right\rangle_\gamma .$$

This gives a finite graph diagnostic of how holonomy ensembles affect shadow localization. No particle mass, physical confinement, or empirical matter field is claimed.

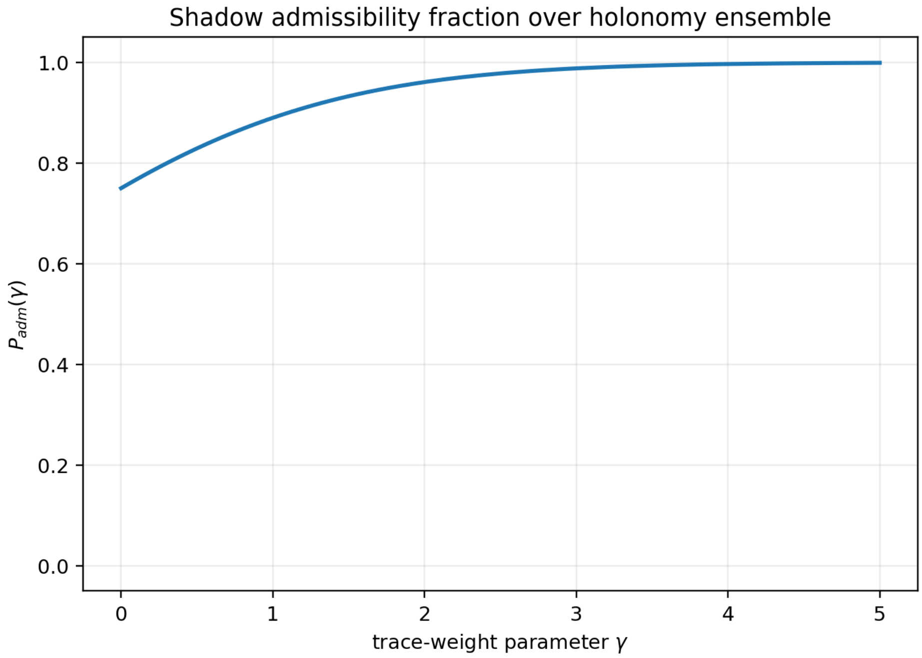

Figure 4: Shadow admissibility fraction over a finite holonomy ensemble. This is a finite graph diagnostic of local shadow admissibility, not a physical matter-confinement theorem.

13 Wilson-loop diagnostics and confinement analogues

For a closed cycle C, define the ensemble expectation

$$\langle \Omega_C \rangle_\gamma = \frac{1}{Z} \sum_{\varepsilon} \Omega_C \exp\left(\gamma_3 \operatorname{Tr}(A_\varepsilon^3)\right).$$

On finite graphs, one can measure whether larger cycles show rapid decay with a chosen notion of enclosed area. This is only a finite graph area-law diagnostic, not a proof of continuum confinement.

Allowed statement:

The ensemble supports finite-graph Wilson-loop diagnostics.

Forbidden statement:

BC-Origin proves physical confinement.

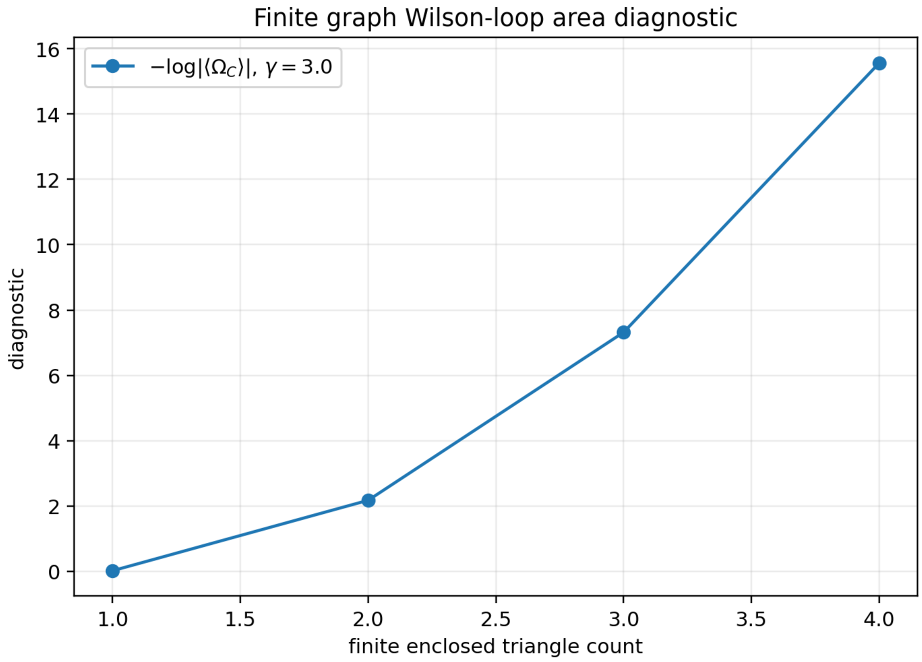

Figure 5: Finite graph area-law diagnostic. The figure shows how Wilson-loop diagnostics can be computed on a finite triangulated fan. No continuum confinement theorem is claimed.

14 Minimal exact examples

14.1 Single triangle

For three vertices and three edges,

$$\mathrm{Tr}(A_\varepsilon^3) = 6\Omega_\triangle K_{12}K_{23}K_{31}.$$

The partition function is

$$Z = 8\cosh(6\gamma K_\triangle),$$

if summing over all edge sign configurations. The expectation is

$$\langle \Omega_\triangle \rangle = \tanh(6\gamma K_\triangle).$$

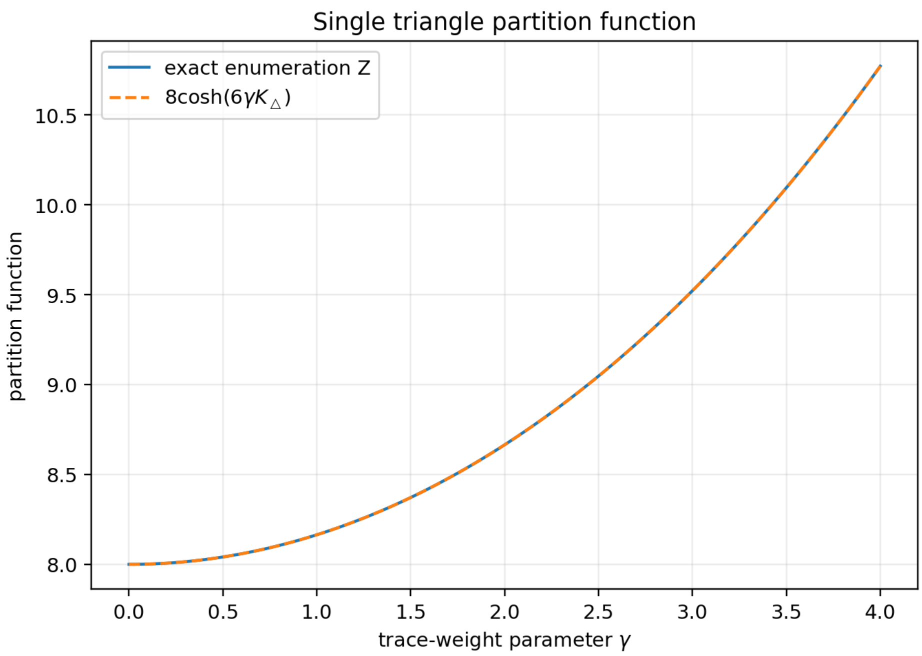

Figure 6: Single triangle partition function. Exact enumeration agrees with the analytic expression $8\cosh(6\gamma K_{\triangle})$.

Figure 7: Single triangle Wilson-cycle expectation. Exact enumeration agrees with $\tanh(6\gamma K_{\triangle})$.

14.2 Two triangles sharing one edge

The ensemble contains two coupled triangle holonomies because the same edge sign participates in both cycles. This is the smallest nontrivial interacting gauge-weight example.

Figure 8: Two triangles sharing one edge. The finite simplicial ensemble couples triangle holonomies through shared edge signs.

14.3 Finite-graph crossover diagnostics

On finite graphs one should speak of crossovers, finite-size susceptibility peaks, or diagnostic changes, not thermodynamic phase transitions. A thermodynamic transition requires a size limit and finite-size scaling.

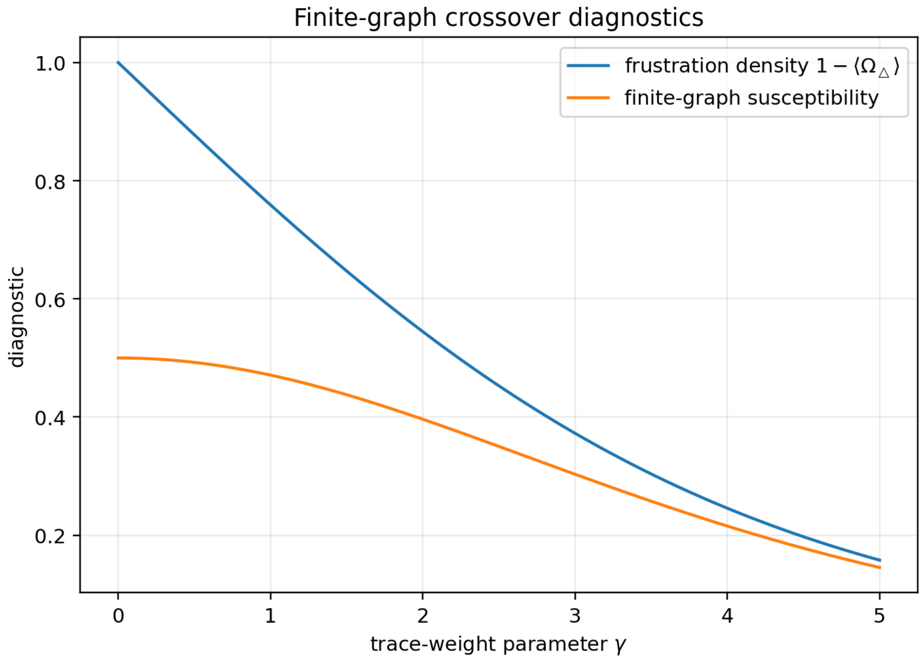

Figure 9: Finite-graph frustration and susceptibility diagnostics. These are not thermodynamic phase-transition claims.

15 Software companion

The software companion consists of a reproducible Python layer and an interactive HTML layer.

Python layer. The Python package performs exact enumeration for small graphs, verifies trace identities, computes finite graph partition functions, measures Wilson-cycle expectations, and computes shadow-admissibility fractions. It also includes an optional Metropolis sampler for larger exploratory graphs, but exact enumeration remains primary for review examples.

HTML layer. The interactive HTML lab visualizes the selected finite graph, edge holonomies, triangle products $\Omega_\triangle$, exact ensemble diagnostics, and shadow admissibility. It is a reader-facing demonstration, not a physical simulator.

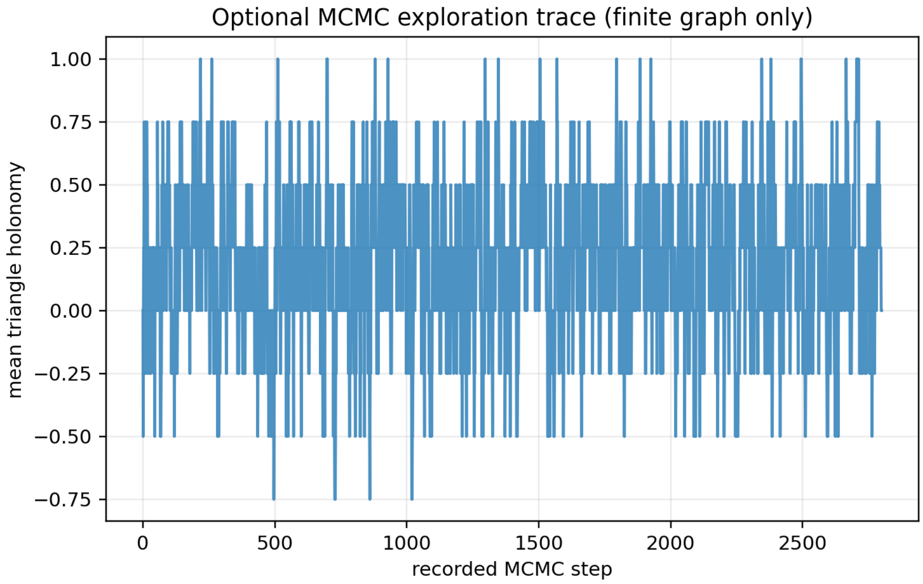

Figure 10: Optional MCMC exploration trace on a larger finite graph. This is a finite graph sampling tool and not a physical QFT simulation.

16 Relation to BC-Origin I–IV

- BC-Origin I: closure-selected shadows.
- BC-Origin II: structural kernels.
- BC-Origin III: non-factorizable edge holonomy.
- BC-Origin IV: lifted phase-flow and horizon events.
- BC-Origin V: spectral traces generate gauge-cycle weights and finite simplicial ensembles.

The present article should not be read as a replacement for the earlier modules. It is a structural bridge from static holonomy to finite statistical gauge-weight ensembles.

17 Controlled claims

The paper claims:

1. Z_2 edge holonomy is naturally represented by signed structural adjacency operators.
2. Closed spectral traces generate closed-walk holonomy products.
3. Triangle Wilson-cycle terms appear in $\mathrm{Tr}(A_\varepsilon^3)$.
4. A finite simplicial gauge-weight ensemble can be defined from spectral trace functionals.
5. Structural kernels induce inhomogeneous Wilson-like cycle weights.
6. Shadow admissibility can be averaged over the finite holonomy ensemble.
7. Finite graph Wilson-loop and admissibility diagnostics provide a controlled confinement-like diagnostic language.

18 Non-claims and limitations

The paper does not claim:

1. derivation of physical lattice quantum field theory;
2. derivation of QCD or Standard Model gauge theory;
3. proof of continuum confinement;
4. empirical spin-glass dynamics;
5. physical path integral over real fields;
6. derivation of matter particles;
7. quantum gravity;
8. continuum limit;
9. physical time dynamics;
10. experimental prediction.

19 Conclusion

BC-Origin V shows that the non-factorizable holonomy introduced in BC-Origin III is not merely a static sign decoration. Once inserted into a structural adjacency operator, it appears automatically in spectral trace cycles. By using these trace functionals as weights in a finite simplicial ensemble, the programme obtains a controlled statistical bridge from shadow holonomy to Wilson-like gauge-cycle weights.

This is not yet physical lattice quantum field theory. It is a finite-dimensional spectral-gauge ensemble generated from BC-Origin structural data. The next natural task is to determine which finite graph confinement-like diagnostics are robust under graph refinement, kernel deformation, and admissibility threshold changes.

A Formal review checklist

Before publication, reviewers should verify:

1. equation numbering and theorem numbering;
2. trace/action distinction;
3. gauge invariance of cycle products;
4. exact single triangle formulas;
5. exact enumeration checks in the software;
6. conservative confinement-language discipline;
7. rendering of all figures and formulas.

B Zenodo metadata draft

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